

James V. Raj

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EDUCATION

University of California, Los Angeles Expected June 2027
Ph.D. in Physics. Advisor: [Mason A. Porter](#). Advanced to candidacy March 2025.
California Polytechnic State University, San Luis Obispo 2021
B.S. in Physics

RESEARCH INTERESTS

Networks that arise from physical systems; hyperuniform point patterns; scale-rich metamaterials; network science; physical learning with local learning rules; topological data analysis and persistent homology; statistical physics.

PUBLICATIONS

Peer-reviewed journal articles

R. Sengupta, M. D. Tikekar, **J. V. Raj**, K. T. Delaney, M. C. Villet, and G. H. Fredrickson, “Phase-field simulations of morphology development in reactive polymer blending,” *J. Rheol.* **67**, 1 (2023).
[doi:10.1122/8.0000523](https://doi.org/10.1122/8.0000523)

Preprints and papers under review

J. V. Raj, X. Sun, C. E. Maher, K. A. Newhall, and M. A. Porter, “Local geometric and transport properties of networks that are generated from hyperuniform point patterns,” [arXiv:2511.21082](#) (2025); in revision at *Phys. Rev. E*. Code: github.com/DMREF-networks/HuPPI-Network-Analysis.

Manuscripts in preparation

J. V. Raj, C. E. Maher, K. A. Newhall, and M. A. Porter, “Size-resolved H_0 persistence as an exact coalescence observable,” in preparation (2026).

J. V. Raj, C. E. Maher, K. A. Newhall, and M. A. Porter, “A directional transport surrogate for stiffness in disordered beam networks,” in preparation (2026).

J. V. Raj, L. Gyllingberg, P. B. Pietrzkowski, M. Stern, and M. A. Porter, “Network structure controls how quickly local learning opens spectral gaps in spring networks,” in preparation (2026).

HONORS, AWARDS, AND FUNDING

Outstanding Teaching Award, Department of Physics and Astronomy, UCLA, for 2022 to 2024
the 2022 to 2023 and 2023 to 2024 academic years

Graduate research support, NSF DMREF award CMMI-2323343 (UCLA) 2024 to present

Future Leaders in Advanced Materials Internship Award 2020

William and Linda Frost Undergraduate Research Award, Cal Poly 2019

PRESENTATIONS

Contributed talks

- “Network structure shapes accessible solutions in physical learning”
Conference on Complex Systems (CCS 2026), Binghamton University,
Binghamton, NY October 2026
- “Topological and transport properties of spatially embedded networks generated
by hyperuniform point patterns”
APS March Meeting, Anaheim, CA March 2025
- “Understanding how the glass transition concentration affects polymer membrane
microstructure”
Future Leaders in Advanced Materials Summer Symposium August 2020

Posters

- “Electrical current characteristics of simulated carbon nanotube network
field-effect transistors”
APS Far West Section Meeting October 2019

RESEARCH EXPERIENCE

- Graduate Student Researcher**, UCLA 2024 to present
Advisor: Mason A. Porter, Department of Mathematics
- Graduate Student Researcher**, UCLA 2022 to 2023
Advisor: Shenshen Wang, Department of Physics and Astronomy
Theoretical biophysics of adaptive immunity and evolving populations.
- Postbaccalaureate Researcher**, Materials Research Laboratory, UC Santa Barbara 2021 to 2022
Advisor: Glenn H. Fredrickson, Department of Chemical Engineering
Designed and ran phase-field simulations of reactive polymer blending under
shear flow on UCSB’s high-performance computing clusters; wrote the Python
simulation-input and ParaView analysis scripts.
- Summer Research Intern** (NSF-funded), UC Santa Barbara 2020
Advisor: Glenn H. Fredrickson
Simulated polymer membrane formation on high-performance computing
clusters; wrote Python tools to quantify pore structure in SEM images and
compare it with simulations.
- Soft Matter Research Assistant**, Cal Poly 2019 to 2021
Advisor: Hilary Jacks
Statistical analysis of memory formation in non-Brownian suspensions under
cyclic shear, including an algorithm to read multiple stored memories from
particle positions.
- Nanoelectronics Research Assistant**, Cal Poly 2018 to 2019
Advisor: Colleen Marlow; includes a 2019 summer research internship
Simulations of carbon nanotube network field-effect transistors: network
morphology, percolation, and sensitivity to electrostatic gating.

TEACHING

Teaching Assistant, Department of Physics and Astronomy, UCLA 2023 to 2025
Upper division: Introduction to Galactic Astronomy (Winter 2023, Winter 2024);
Thermal Physics (Spring 2023); Mathematical Methods (Fall 2023)
Lower division (Physics for Scientists and Engineers sequence): Mechanics (Winter 2025); Oscillations, Waves, Electric and Magnetic Fields (Fall 2024, Spring 2025)

MENTORING

Graduate mentor, UCLA Marginalized Identities in Physics & Astronomy (MiPA): 2023 to 2025
mentored eight undergraduate students

Graduate student mentor: mentored two graduate students (2023 to 2024 and 2024 to 2025 academic years) 2023 to 2025

SERVICE

Assisted with a referee report for *Interfaces and Free Boundaries* 2025

Administrator of the Porter research group's shared Linux compute server 2025 to present

Vice President, Sigma Pi Sigma (physics honor society), Cal Poly 2020 to 2021

COMPUTATIONAL SKILLS

Languages: Python, Bash, $\text{\LaTeX}$, JavaScript/HTML/CSS, Rust (sparse numerical kernels called from Python)

Scientific computing: NumPy, SciPy (sparse linear algebra, eigensolvers), NetworkX, pandas, Matplotlib, Plotly, ParaView; parallel computing with multiprocessing and joblib; GUDHI, Ripser, and persim for persistent homology

Machine learning: PyTorch (CUDA, `torch.compile`, differentiable optimization); LoRA fine-tuning of diffusion models; local deployment of open-weight LLMs with llama.cpp (quantization, speculative decoding, mixture-of-experts offloading)

Systems: batch jobs on high-performance computing clusters; Linux server administration (user accounts, systemd services, networking) for a 96-core, 1 TB RAM GPU server; Git and GitHub; pytest